

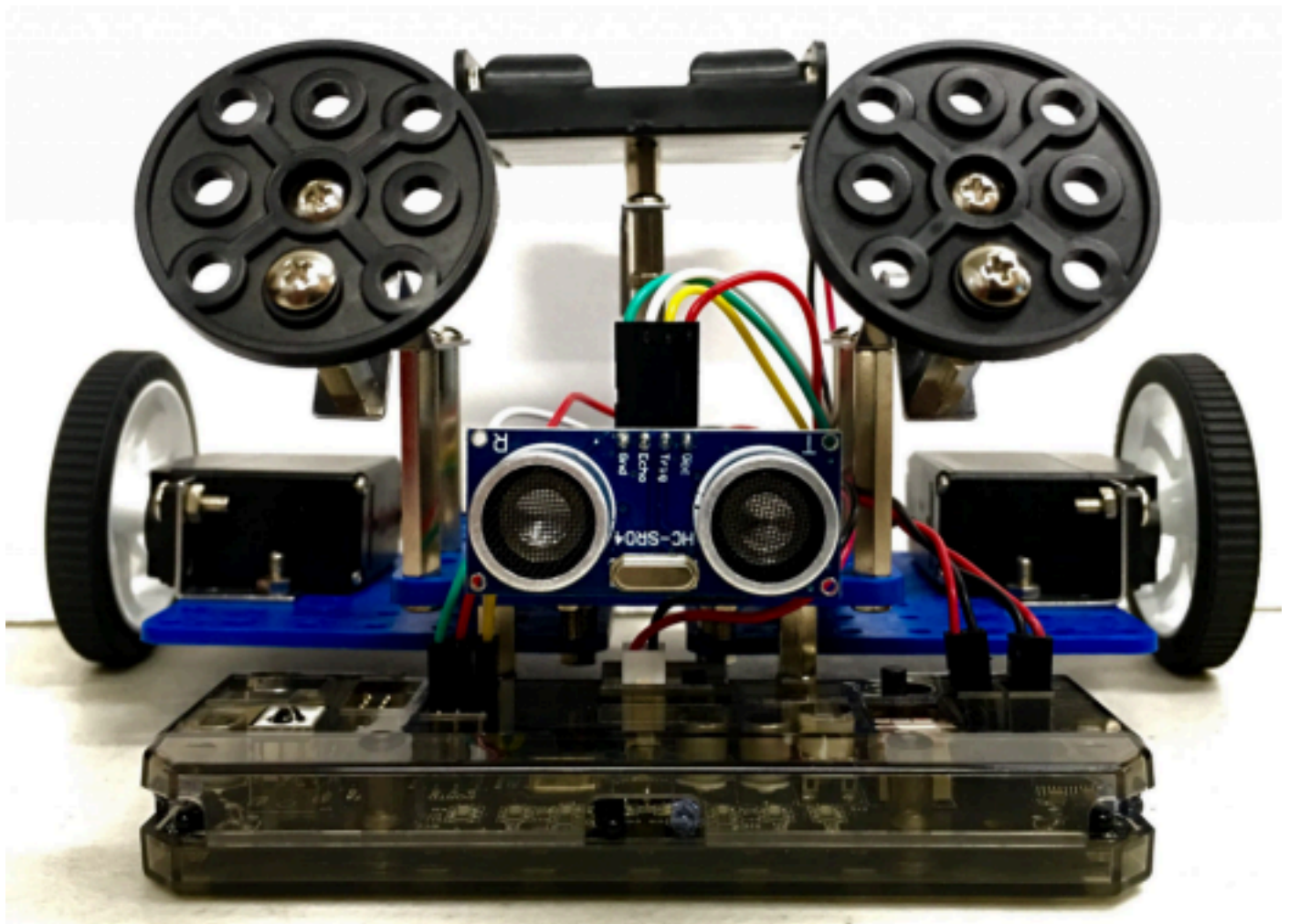
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Macau Instituto Salesiano DAT project

Sumo Robot

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MENU

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1 ABSTRACT:

1.1 PROJECT GOAL

This project's goal is to develop a small sumo-like robot that is capable of self-identifying and evicting an opponent from a rectangular wooden arena. Picture a self-sufficient sumo wrestler who works in a field of robotics using infrared sensors, autonomous navigation, and other vertebrates from the animal kingdom as inspiration. Designed robots with traits of quick reflexes, strategic foresight, and precision task execution. Key responsibilities include authoring control routines for autonomous navigation and pushing, integrating infrared sensors for opponent detection, designing the mechanical components, including wheels, the pushing apparatus, and the chassis, and iteratively refining performance to gain a competitive edge in the arena. Creating a robust and agile micro sumo robot is the final aim.

1.2 SENSOR INTRODUCTION

The project's key technologies include seven infrared sensors. Three of which are placed at the top of the robot for maximum opponent tracking, and four are used to ensure the robot remains within the designated arena. A microcontroller uses the calculated data from the sensors regarding the opponent's distance and their proximity to the edge of the arena, and makes proportional adjustments to movement strategies in real time. All processing is performed within the unit, which allows for highly optimized responsiveness and self-governance, providing unparalleled agility. This optimizes total control and immediacy, as no additional data transit is conducted.

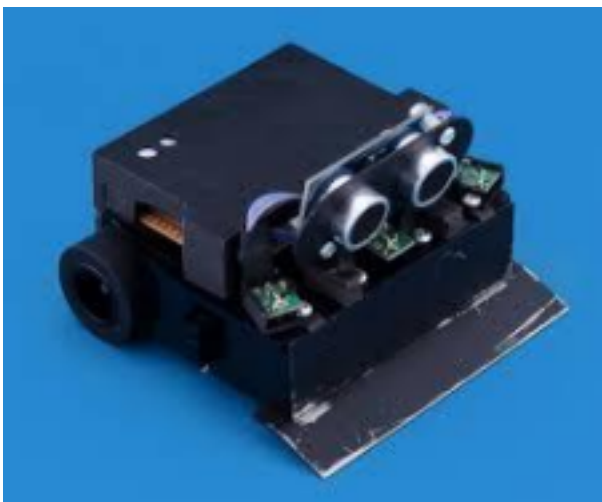


Fig. 1 Sumo Robot

2. BACKGROUND:

A sumo robot is an autonomous mobile robot designed to compete head-to-head against an opponent within a circular space. The primary objective is to either force the opposing robot out of the ring or immobilize it.



Fig 2 dohyo

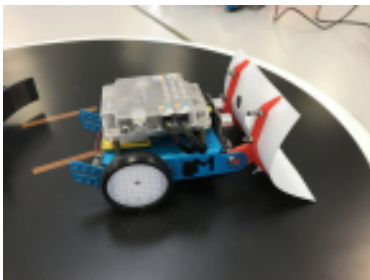
Robot sumo wrestling originated in Japan, emerging from a blend of robotics enthusiasm, engineering challenges, and cultural reverence for traditional sumo. Early informal competitions at universities let students showcase engineering skills. By 1989, organized events like the All-Japan Robot-Sumo Tournament standardized rules and spurred technical advancements, elevating the sport globally. Today, competitions thrive worldwide, uniting diverse innovators.

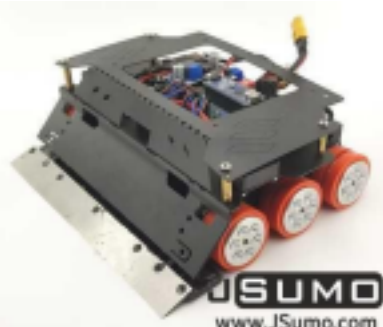
A sumo robot's goal is to push or immobilize its opponent to force a ring-out or disablement. It must adhere to strict size limits (200mm × 200mm × 200mm) and a maximum weight of 3kg. Equipped with sensors to detect opponents and the ring's edge, it strategically maneuvers to eject rivals from the dohyo (arena). Weapons and intentional damage are prohibited. Victory is awarded to the robot remaining in the ring after a match. If time expires, the lighter robot wins.

3. DESIGN

3.1 EXAMPLES

- Here are 4 examples of sumo robots to know their advantages and the disadvantages to compare what are our good reference examples for us to make our own sumo robot.

Example1  Fig 3 MBOT	Advantages: • The sumo robot is built using MBOT • It is easy to program since it supports block-based coding. • Only requires attaching a front wing, making it simple to build and repair. • No additional design or CAD modeling is required. • Uses a ground sensor to detect the surface color, preventing the robot from leaving the sumo ring	Disadvantages: • MBOT is expensive, making it a less budget-friendly option. • If any compounds have any failure, replacement is required, which takes time and additional cost. • The default motors are not strong enough to push opponents effectively. • Upgrading to stronger motors may be necessary, adding to the cost.
examples2:  Fig 4 sources: Black Mamba	• The sumo robot uses an ultrasonic sensor to detect others. • An IR proximity sensor assists the ultrasonic sensor in detection. • It utilizes 50:1 HP micro metal gearmotors for a good balance of speed and torque suitable for mini sumo. • Weight is added to bring the robot close to the maximum weight limit for the race, as calculating weight is not necessary to determine the winner.	• Incorrect material choices can lead to wasted money and time. • Building the car independently can be difficult. • Troubleshooting and fixing may take significant effort. • If the car is too heavy, it may reduce speed and maneuverability, increasing the risk of losing the race.

example3:  Fig 5	• Includes essential components like the BB1 chassis, XMotion controller, and various sensors for effective operation. • Choice of motor speeds (250 RPM or 500 RPM) allows for customized performance based on race requirements. • Electron wheels directly attach to the 25mm diameter motors, ensuring ease of assembly. • Multiple opponent and line sensors enhance detection and navigation capabilities.	• Building the robot from individual components may be challenging for beginners. • Motors rated for 12V can be overvoltage to 18V, which may lead to overheating or damage if not managed carefully. • Proper weight management is crucial; too much weight can hinder performance and increase the risk of losing races. • Requires communication for specific motor speed options, which may lead to delays if not specified before shipment.
example4:  Fig 6	• It has strong power to push other cars out of the way. • It can add extra components to make it more useful. • It has a strong front wing that can push others easily.	• It has six tires, which may make the car overweight. • The materials for this car are very expensive, costing around \$2,250, so it is not suitable for us to use. • The car requires a lot of time for testing and needs multiple trials.

We would like to prefer Example 2 for our sumo robot project because it demonstrates several key advantages. Example 2 is designed to move smoothly, which enhances its agility during competition. Additionally, this robot is equipped with advanced sensors that allow it to detect other robots effectively. This capability enables it to identify opponents quickly and launch attacks with precision. Overall, the features of Example 2 make it a highly effective model for our project, as it combines smooth movement with efficient targeting.

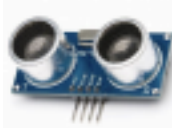
3.2 MOTOR

Here are the compare of the 4 motors to let us know which motor is the most suitable for our project

MOTOR	Tamiya 70093	L Type 520 encoder motor	520 DC reduction motor	310 DC reduction motor
PHOTO	 Fig 7	 Fig 8	 Fig 9	 Fig 10
GEAR RATIO	High speed 16.6:1	1:40	1:56	1:20
Rated Voltage	3VDC	12V	12V	7.4V
RPM	12300rpm	12000 rpm	11000-12000 rpm	9000 rpm
output rotational speed	740 rpm	300 rpm	205-550 rpm	450 rpm
Output torque	0.6 kg.cm	10kg.cm	3.1kg/cm~8.3kg/cm 0	0.05kg/cm~11.0kg/cm

The L Type 520 Encoder Motor is an outstanding choice for demanding applications, combining remarkable features that enhance performance and reliability. With a torque output of up to 10 kg.cm, it adeptly handles heavy loads, making it ideal for high-intensity tasks. Its capability to reach a maximum speed of 12,000 rpm ensures rapid response and efficient operation, while the built-in encoder provides precise control over speed and position, significantly improving automated systems. Compact and easy to integrate into various projects, this motor strikes a perfect balance between efficiency and power consumption, allowing for continuous operation without overheating. In summary, the L Type 520 Encoder Motor is an ideal solution for a diverse range of applications.

3.3 SENSORS

	Ultrasonic Sensor	IRsensor	IRsensor	Laser sensor	Micro Switch Sensor
Photo	 Fig 11	 Fig 12	 Fig 13	 Fig 14	 Fig 15
Model Name	HC-SR04 Ultrasonic Distance Module	TCRT 5000 infrared reflective sensor	IR Obstacle Avoidance Sensor	TF Mini LiDAR	Micro Switch Sensor
Operating Principle	Measures distance by calculating the time between transmitting and receiving ultrasonic waves.	Determine the distance and presence of an object by measuring the difference in intensity between transmitted and received infrared light	Determine the presence of an object by measuring the difference in intensity between transmitted and received infrared light	Light is emitted from the light source and reflected by objects in the scene. The system's receiver detects the reflected light and uses time-of-flight (TOF) to map the distances of objects in the scene.	An electrical switch triggered by very small physical forces, widely used in household appliances, machinery, industrial control, and more
Input Signal	working range message	1/0	UART	working range value	0/1
Specifications	Size: 45mm x 20mm x 15mm Working Voltage: 5V	Size: 32*14*10 (mm) Voltage: 3.3V-5V	Size: 32* 14 (mm) Voltage 3.3V-5V	Size: 42*15*16 (mm) Voltage : 5V	Size: 30*20*8 (mm) voltage : 5V
Working Range	Detection Range 2-400cm	Detection Range 1-10cm	Detection Range 2-30cm	Detection Range 0.3m-12m	Physical contact trigger

Here are the four sensors that compare to know which is best for our sumo robot. After comparing the available sensors, we have decided to choose the IR sensors for our sumo robot. It is the easiest to use and program, and it is the most affordable option. This simplicity allows us to focus on other aspects of our robot's design.

3.4 DESIGN TARGETS

Dimensions:

- The robot must fit within a specified size limit, typically a square of 200mm*200mm*200mm.

Weight:

- The maximum weight for the robot is 3kg

Microcontroller:

- A common choice for the microcontroller is the Arduino

Sensor Types:

- We need to use the sensor to keep us on track

Number of Sensors:

- It is advisable to incorporate at least 3 to 6 sensors to ensure comprehensive coverage for detecting edges and opponents

Voltage:

- The robot should operate on a voltage range of 6V

Motors and Gearbox:

- High-torque DC motors are recommended for movement, with a typical speed range of 100 to 300 RPM. The motors should be paired with a suitable gearbox to maximize torque and control

Body Material:

- The body of the robot can be constructed from lightweight materials such as plastics or aluminum alloys. These materials provide durability while keeping the overall weight low

Shovel Design:

- The shovel or pushing mechanism can be designed with a flat or angled shape to effectively lift and push opponents out of the ring. It is crucial to ensure that the shovel is sturdy and does not hinder the robot's movement.

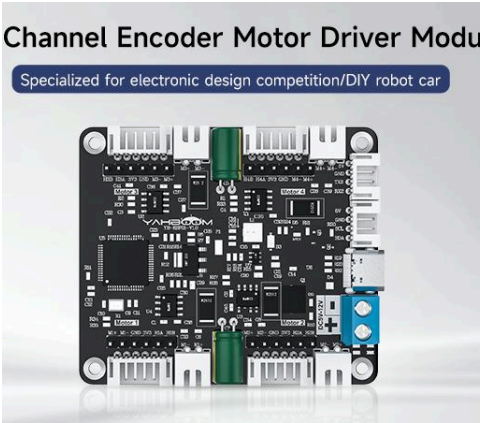
4. TECHNICAL SPECIFICATIONS

4.1 MOTOR CONTROL

Integrating IIC in a sumo robot motor can enhance communication and control. One of the key benefits is the ability to connect multiple sensors, such as distance sensors or gyroscopes, to the same IIC bus. This allows the robot to gather data from various sources efficiently. Additionally, using IIC simplifies wiring, reducing the number of wires needed for communication and making the robot's construction cleaner and easier to manage.

IIC also enables real-time control by facilitating quick data exchange between the microcontroller and motor drivers, allowing for responsive motor control based on sensor inputs. The protocol's flexible configuration allows for easy addition or removal of devices on the IIC bus, making design adjustments straightforward. Furthermore, IIC is ideal for battery-powered robots, helping to conserve energy, which is crucial in competitive environments like sumo wrestling. By leveraging IIC, your sumo robot can achieve better performance and adaptability in the arena.

And here I will introduce the motor shield that we used in this project, it costs ¥58, it is made by Yaboom, and the motor is also made by this company

Four-channel encoder motor drive module. The four-channel encoder motor drive module integrates a high-performance single-chip coprocessor and can communicate with MSPM0 Multiple controllers, such as STM32, Raspberry Pi, and Jetson, can be seamlessly connected to simplify the driving process. Only four connecting wires are needed to connect to the main control unit. The efficient communication of the elements makes it easy to control the motor and obtain encoder data, reducing the number of wires and the difficulty of operation. Also supports drivers Most of the Hall encoder TT motors, 520/310, and other DC reduction motors on the market.	Channel Encoder Motor Driver Modu Specialized for electronic design competition/DIY robot car  Fig 16
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4.2 SENSOR CONTROL

Infrared (IR) sensors detect objects and obstacles by emitting infrared light and analyzing the reflections. The process begins with the sensor emitting IR light, typically from an LED, which is invisible to the human eye. When this emitted light encounters an object, it reflects toward the sensor. The sensor is equipped with a photodetector that detects this reflected IR light.

By measuring the intensity of the reflected light, the sensor can determine the distance to the object. In some cases, the sensor measures the time it takes for the light to return, allowing for precise distance calculations. This capability makes IR sensors particularly useful in robotics, where they are employed for obstacle detection and avoidance.

Additionally, IR sensors find applications in security systems to detect intruders and in automotive technology for parking assistance and collision avoidance. Their effectiveness and cost-efficiency make IR sensors a popular choice for various object detection applications.

We need two ground sensors to keep the car stable, specifically using the Dohyo. Currently, we have six infrared (IR) sensors to detect other vehicles. However, IR sensors are fragile, and we have two that are broken, so we need to reduce our number from six to four. We can adjust the value of the resistor to change the distance the IR sensors can scan. On the other hand, if the sensors are positioned too high, they may not detect other cars that are too low. Additionally, if the sensors are placed too far back, they might detect themselves, leading to inaccurate readings.

4.3 PROGRAMMING

1. Set the motor type. This motor shield supports multiple motors, so we need to specify which motor we are using.

- `FUNCTION Set_motor_type()`
- `// Set to L type 520 motor`
- `END FUNCTION`

2. Set the pulse phase

- `FUNCTION Configure_Motor()`
- `Set_Pulse_Phase(30) // Configure the reduction ratio`
- `// Additional configuration if needed`
- `END FUNCTION`

3. Control the speed (allow us to control it with 8 different settings).

```
FUNCTION control_speed(m1, m2, m3, m4)
    DEFINE speed AS ARRAY[8] OF uint8
    speed[0] = (m1 >> 8) AND 0xff
    speed[1] = m1 AND 0xff
    speed[2] = (m2 >> 8) AND 0xff
    speed[3] = m2 AND 0xff
    speed[4] = (m3 >> 8) AND 0xff
    speed[5] = m3 AND 0xff
```

```

    speed[6] = (m4 >> 8) AND 0xff
    speed[7] = m4 AND 0xff
    CALL i2cWrite(Motor_model_ADDR, SPEED_Control_REG,
    8, speed)

```

5. DESIGN SPECIFICATIONS

5.1 BODY DESIGN

- The tires have a diameter of 67.5 mm.
- The robot's wheelbase measures 140 mm.
- It is equipped with a total of seven sensors.
- We utilize integrated molding for the shell and front wing.
- These features contribute to making it less likely to be pushed off the track.
- The dimensions were determined based on the size of the motor and the motor shield.

5.2 CAD

- Our CAD makes a special design, is that we make Integrated molding in our shell and the front wing, so that we can easily design and connect it to the robot

The problem we faced :

1. Using the wrong scale when drawing
2. Lack of space on the base
3. No space for a ground sensor because the base is too low to the ground
4. The motor is too big, which causes the battery box to have no space to put

5.3 VERSIONS

- The first version of the sumo robot had a lot of problems, such as it did not have enough space for our new motor, so we needed to draw a new one, and also we changed a lot of things, such as the shell of the robot, and we created a better space for the motor for the new L-shaped motor

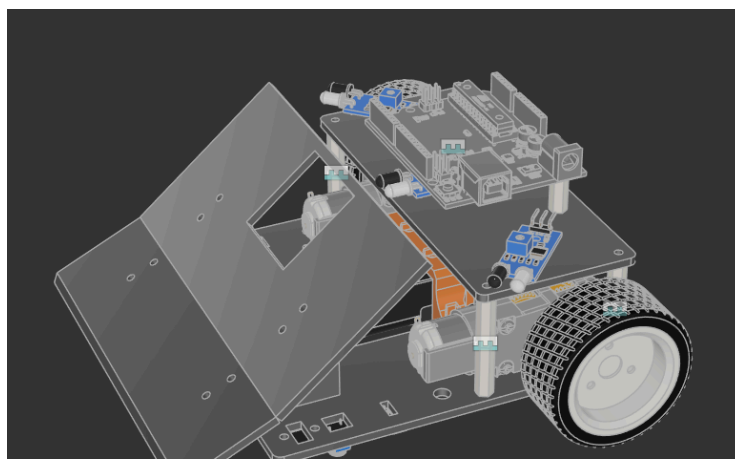


Fig 17 First version of the sumo robot

- So we change to the new version of the sumo robot that is all drawn by Elwin, and we fix all the problems that we see in the Fusion

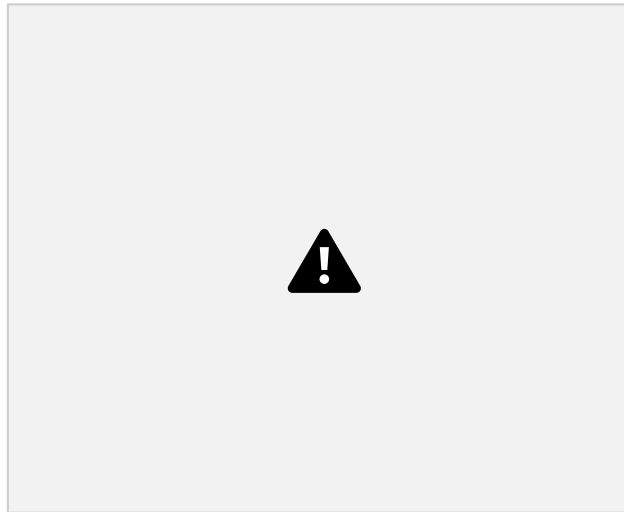


Fig 18 The last version of our sumo robot

- But we face a big problem, which is that the motor is too close to the battery box, making the motor unable to turn, so we spend some time polishing the battery box to make space for the motor to move
- When we first start to test our car, and our front wing is broken, and we have no opportunity to 3D print a new one, so we stick some adhesive tape to add some support to the front wing to prevent Secondary injury to it, so we fix our car and have a good race

5.4 MATERIAL

- We chose 3D printing for the front wing and body parts of our sumo robot.
 - 3D printing is the most convenient option for our assembly, and we believe our car would be too heavy if we chose an acrylic sheet for the front wing.
- When we were making our car, the shell was integrated through molding. However, when connecting it to the other parts, we encountered issues such as screws becoming loose. We had to remove the entire shell to retrieve the screws, which wasted a lot of time.



Fig.19 bambu 3D printing machine

6. RESULT

6.1 COMPETITION

- In this competition, we had the power to push others off the track. However, when we reached the IR sensors at the end, we couldn't use it, which confused us. Nevertheless, we finished in 4th place, which is still a good result. Although we were a bit upset about not making it to the finals, we did our best, so there's nothing to be disappointed about. Good job!
- When we race, our car is very strong and can easily push others out. It is suitable for races with more than three cars. However, when we need to race against other cars one by one, our car struggles to locate them, which wastes a lot of time and causes us to miss out on victory.
- **Racing experience**
 - **Gordon:** Participating in the sumo robot competition was an exciting experience. Our robot was strong and designed to push opponents out of the ring. While it performed well in races with many cars, it struggled to scan for opponents in one-on-one matches, causing us to lose time. We finished in 4th place overall.
 - **Gordon:** We learned that we need to trust our teammates and work together to achieve better results. I feel upset about spending so much time and money on this project, only to end up in 4th place. I had hoped we could finish in 2nd place or even win the championship. Unfortunately, our mistakes led to our loss. However, the overall situation has been established, so we can only accept it.
 - **Elwin:** At first, I was so proud of our sumo robot that I was confident we could finish in the top three. However, after our first loss, I began to feel nervous and even blamed myself. During the final match, my anxiety spiked when I realized our robot had technical issues. When I discovered the reason for our defeat, I felt furious—but that anger didn't last. The competition is over, and frustration won't change the outcome. All I can do now is accept it and focus on improving for our next project.

7. CONCLUSION

Dimensions:

- Our sumo robot is within the size limit to confirm that we can race

Weight:

- Our car is the heaviest of all, but we are still within the limit

Microcontroller:

- We use Arduino to control the board and the motor shield

Sensor Types:

- We can only use the edge sensor to help make our car stay in
- The IR sensors are not all used in the race

Motors and Gearbox:

- We use the new motor that can have a stronger power and can give back information on how fast the motor moves to the computer. This is a strong function

Body Material:

- The body materials that we used are plastic that is 3D printed
- The plastic is not strong enough for a big race if other cars are very strong, they may destroy our front wing

Shovel Design:

- At last, the front wing is very close to the ground to give us a strong ground effect to push others off the track and keep us on the track

Future

- In the future, I would like to try to buy a motor that is cheaper and lighter so that our car can have strong power and can win the race
- We can also try to make the car greener and faster; this will be our target,

Teamworks

- We believe that we have good teamwork, and we try our best to create a great car for racing.
- Thanks to Elwin for his hard work in programming and CAD.
- Also, thanks to my team for helping with programming, fixing the car, and contributing to this nice project. We all feel happy about this project, and we hope to work together again

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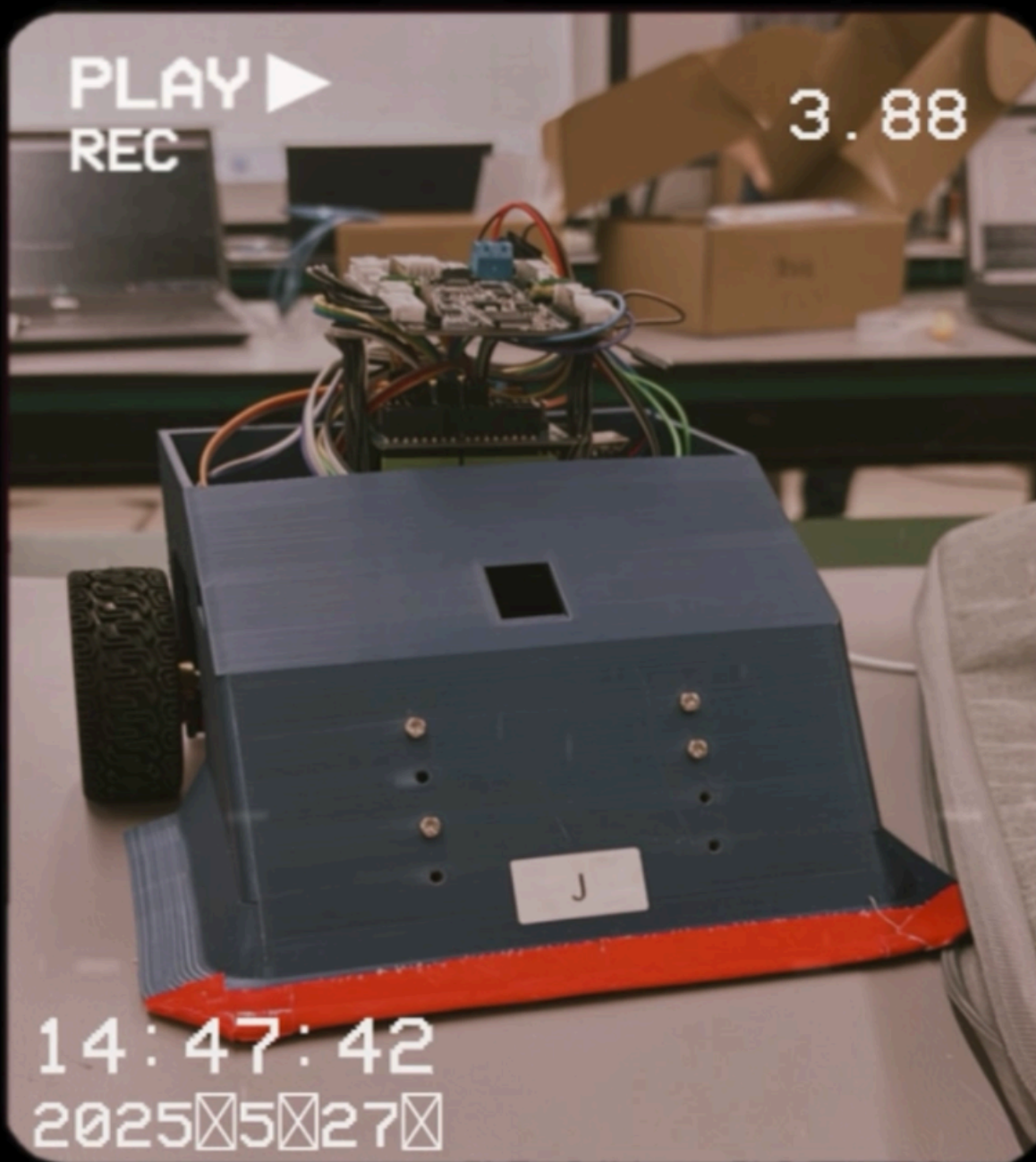
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力量速度兼備